

20 Newsgroups Text Classification with Word2Vec, TF-IDF, BERT, MLM, and Llama-3 + QLoRA

*This work is submitted as part of the final project submission for the NALAPRO course.

Nilabh Pandey

School of Computer Science and Information Technology

Lucerne University of Applied Sciences and Arts

Lucerne, Switzerland

nilabh.pandey@stud.hslu.ch

Abstract—Topic classification of short forum text remains a non-trivial benchmark in natural language processing. In this project I study the 20 Newsgroups corpus and approach the same task with four different families of methods, with the aim of quantifying how much each modelling step actually contributes on a shared benchmark. The progression begins with shallow lexical features, namely averaged Word2Vec embeddings and TF-IDF vectors, applied to a small feed-forward network. It continues with supervised fine-tuning of BERT and, after that, with a variant in which BERT is first adapted to the corpus through masked language modelling and only then fine-tuned. Finally, the eight-billion-parameter Llama-3 model is evaluated in two regimes: through zero-shot and few-shot prompting, and through parameter-efficient fine-tuning with QLoRA. The objective is not to identify a single winner, but to understand the relative contribution of each adaptation strategy. The QLoRA-fine-tuned Llama-3 produces the highest measured accuracy at approximately 78%, although this figure is obtained on a 200-post evaluation sample and is therefore not strictly comparable to the BERT models, which were scored on the full 7,532-document test set and reached around 71%. The combined classical representation reaches 70%, while prompting alone, without any task-specific training, lags substantially at about 51%. Two complementary findings emerge: domain-adaptive masked pre-training yields a small but reliable improvement for BERT, and for the large language model the structure of the prompt has a stronger effect on accuracy than the number of in-context examples shown to it. All training runs are logged to Weights and Biases to support inspection and reproducibility.

Index Terms—text classification, 20 Newsgroups, Word2Vec, TF-IDF, BERT, masked language modeling, Llama-3, QLoRA

I. INTRODUCTION

Assigning a forum post to its corresponding topic is a long-standing problem in natural language processing and a well-established benchmark for evaluating text-representation methods. Manual labelling is straightforward at small scale but does not generalise to thousands of documents, which motivates automated approaches. The 20 Newsgroups corpus has remained a standard testbed for such methods over many years on account of its size, its topical breadth, and the presence of closely related categories.

The corpus is, however, harder than its size alone would suggest. It comprises twenty topics, several of which are thematically close to one another: three concern religion,

three concern political topics, and a further group concerns computing. A post arguing a religious point can be lexically very similar to a post arguing a political one. Since the twenty classes are approximately balanced, a random classifier attains only about five percent accuracy, and any meaningful gain must therefore arise from the lexical or contextual signal present in the posts themselves.

The dominant approach to this problem has changed substantially over time. Early systems represented each document as a vector of weighted term counts, typically TF-IDF, and trained a simple classifier on top. Distributed word vectors such as Word2Vec subsequently introduced a notion of distributional semantics, placing words used in similar contexts close to one another in vector space. Pre-trained transformer encoders such as BERT then offered fine-tunable bidirectional context, and more recent instruction-tuned large language models such as Llama-3 can often address a task directly from a natural-language prompt, with little or no task-specific training.

Rather than selecting a single method, I evaluate all of them on the same split of the same dataset, under a common protocol, in order to make the comparison as informative as possible. The aim is to quantify how much each new modelling step actually contributes, and to interpret the results rather than merely report numbers. The contributions of this work are threefold. First, I provide a controlled, side-by-side comparison of classical features, fine-tuned BERT, domain-adapted BERT, and Llama-3 prompting on the 20 Newsgroups benchmark. Second, I include an optional QLoRA experiment that fine-tunes Llama-3 in a parameter-efficient manner and yields the highest measured accuracy in the study, albeit on a smaller evaluation sample than the other methods. Third, I draw a set of practical conclusions, including the measurable benefit of domain adaptation, the strong effect of prompt formatting, and the residual gap between prompting and fine-tuning at this dataset scale.

II. TOOLS AND REPRODUCIBILITY

All experiments, with their hyper-parameters, training curves, and final metrics, were logged to Weights and Biases to

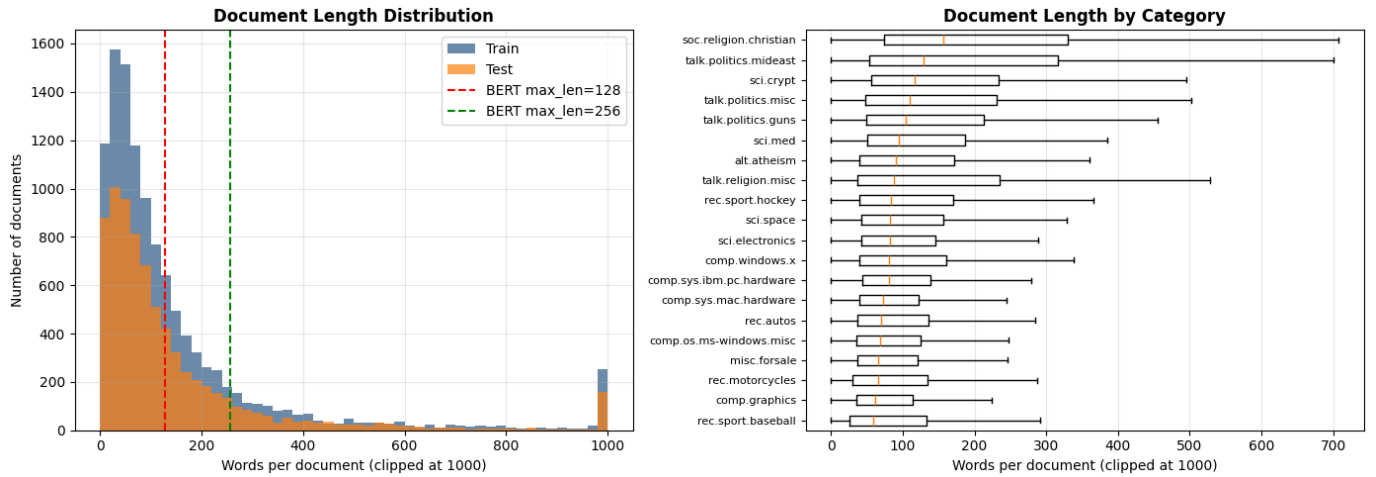


Fig. 1. Document length. Left: the length distribution is strongly right-skewed, with the BERT cut-offs of 128 and 256 tokens marked. Right: length spread by class. Most posts are short, but a few are very long.

support inspection and reproducibility. The project dashboard is available at:

<https://wandb.ai/nilabhpandey7-hslu/nalapro-project>

The complete code and documentation are maintained in the GitHub repository referenced in the Code and Data Availability section. The implementation relies on Python and PyTorch for all models, the Hugging Face Transformers and PEFT libraries for BERT and Llama-3, the `bitsandbytes` library for 4-bit quantisation, scikit-learn for the dataset and the evaluation metrics, Gensim for training Word2Vec, and Matplotlib and Seaborn for the figures. Experiment tracking was handled through Weights and Biases, and the more demanding training runs were performed on Google Colab using T4 and A100 GPUs. Generative AI tools were used as an aid for coding and writing; this is described in detail in the Generative AI Declaration.

III. BACKGROUND AND RELATED WORK

TF-IDF remains a competitive baseline for topic classification and is the canonical frequency-based representation discussed in the course material [11], [12]. It assigns high weight to terms that occur often within a document but rarely across the corpus, which emphasises topic-bearing vocabulary. Word2Vec [2] adopts a complementary, prediction-based view: it learns a dense vector for each word from the words that surround it, so that words used in similar contexts end up close in vector space. A document representation can then be obtained by averaging the vectors of its constituent words.

BERT [3] is a transformer encoder pre-trained on large corpora with the masked-language-modelling objective, in which a fraction of the tokens is hidden and the model learns to predict them from context [14]. Adapting BERT to a new task is performed by fine-tuning, that is, by continuing to update all of its weights on the labelled data for the target task. A related and well-established refinement is to continue pre-training the

model on in-domain text before fine-tuning, which has been shown to produce consistent if often modest improvements [4].

Instruction-tuned large language models such as Llama-3 [6] can be applied without any task-specific training, by simply describing the task in a natural-language prompt. A small number of labelled examples can additionally be inserted into the prompt to enable few-shot learning [5]. The skip-gram and CBOW formulations underlying Word2Vec, together with the few-shot paradigm, are also covered in the course material [13], [15]. When task-specific adaptation is desired, full fine-tuning of a multi-billion-parameter model is computationally expensive. LoRA [7] alleviates this cost by freezing the model and training only small low-rank adapter matrices, and QLoRA [8] reduces the memory footprint further by storing the frozen base model in 4-bit form, enabling fine-tuning of an eight-billion-parameter model on a single GPU.

IV. DATASET, PREPROCESSING, AND EXPLORATORY ANALYSIS

The experiments are conducted on the 20 Newsgroups corpus as distributed by scikit-learn [1]. The corpus contains 18,846 documents spanning 20 topics, of which scikit-learn supplies 11,314 for training and 7,532 for testing. I keep the official test split untouched and partition the training portion into 9,051 documents for training and 2,263 for validation, preserving the per-class proportions in each subset. For the domain-adaptation stage in Task 3, an additional unlabelled corpus of 17,886 in-domain documents is used solely for masked language modelling.

Before any modelling decisions were made, the dataset itself was inspected, because several characteristics of the data have a direct bearing on the choices that follow.

Figure 2 reports the per-class document counts. The classes are approximately balanced, with most topics containing between roughly 480 and 600 training documents;

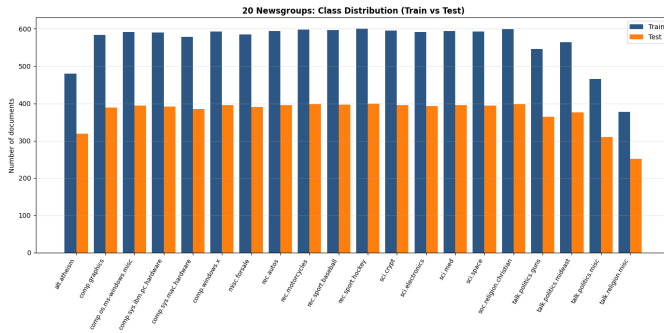


Fig. 2. Number of documents per class in the training and test portions. The classes are close to balanced, so no class weighting is needed.

`talk.religion.misc` and `alt.atheism` are the smallest. Given this balance, no class-weighting scheme is required, and the random baseline corresponds to approximately five percent accuracy. Useful classification performance must therefore be derived from the lexical and contextual content of the posts themselves rather than from any class-frequency artefact.

Document length varies considerably across the corpus, as shown in Figure 1. The mean length is approximately 186 words, whereas the median is only 83, indicating a strongly right-skewed distribution with a long tail; the longest document exceeds 11,000 words. This is directly relevant to BERT, which operates on a fixed-length token window. After tokenisation, approximately 34% of posts exceed 128 tokens and approximately 14% exceed 256. I therefore set the BERT maximum sequence length to 256, which preserves the vast majority of each document while a cut-off of 128 would truncate a substantial fraction of the corpus.

Prior to training, the headers, footers, and quoted reply blocks are stripped from every post. Although superficially minor, this step is important: these regions frequently contain sender identifiers, e-mail addresses, and newsgroup names, all of which leak the label directly. Their removal prevents models from exploiting metadata artefacts and forces them to learn from the actual textual content of each post.

Subsequent preprocessing is method-dependent and reflects a deliberate distinction between the classical and the transformer-based pipelines. For the classical models in Task 1, the text is aggressively cleaned: it is lowercased, and stopwords, numerals, URLs, and e-mail addresses are removed, in line with a standard NLP pipeline [9]. A small network with no linguistic priors benefits from this normalisation. For BERT and Llama-3 the opposite policy is applied: the text is left close to its natural form and handed to the model’s own subword tokeniser [10], since these models were pre-trained on naturally occurring text and overly aggressive cleaning would discard signal that they are equipped to exploit.

The vocabulary is large but extremely uneven. The training set comprises approximately 1.6 million tokens drawn from a vocabulary of 46,733 unique types, of which 42.2% are hapax legomena, that is, words that occur exactly once. Figure 3

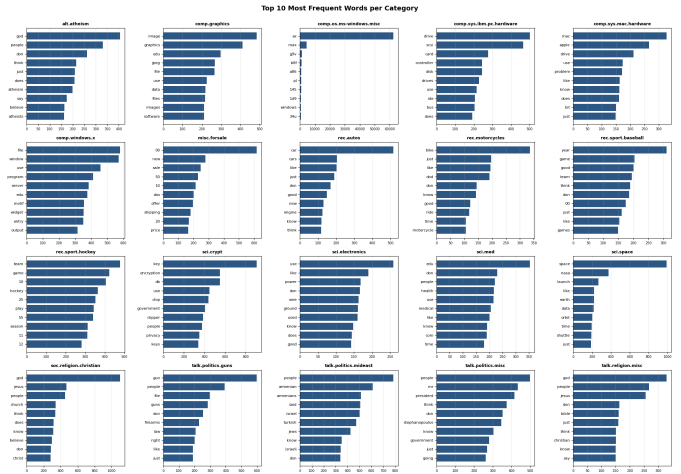


Fig. 3. Word frequency against rank on log-log axes. The near-straight line is the classic shape of Zipf’s law for natural language.

plots word frequency against rank on log-log axes and exhibits the approximately linear shape that is characteristic of Zipf’s law for natural language [12]. The long tail of rare words motivates the use of a minimum-document-frequency cut-off for the TF-IDF vocabulary, which prevents the network from being exposed to many thousands of low-frequency features.

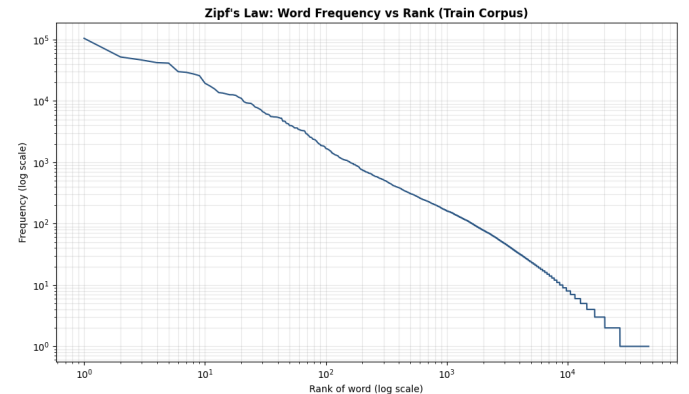


Fig. 4. The ten most frequent words in each class after stopwords removal. Many classes have a clear signature; a few are shaped by quirks of the data.

Figure 4 reports the ten most frequent terms within each class after stopwords removal. Many classes display a clearly recognisable lexical signature: `god` and `atheism` for the religion groups, `key` and `encryption` for `sci.crypt`, `car` for `rec.autos`, `gun` for `talk.politics.guns`, and `space` and `nasa` for `sci.space`. These distinctive keywords are precisely the signal that TF-IDF amplifies, and they account for the relative ease with which topically distinct classes can be separated. The figure also exposes a data artefact: `comp.os.ms-windows.misc` is dominated by the token `ax`, which originates from a single encoded binary post rather than from genuine discussion, indicating that the corpus is not perfectly clean.

Figure 5 quantifies the vocabulary similarity between the-

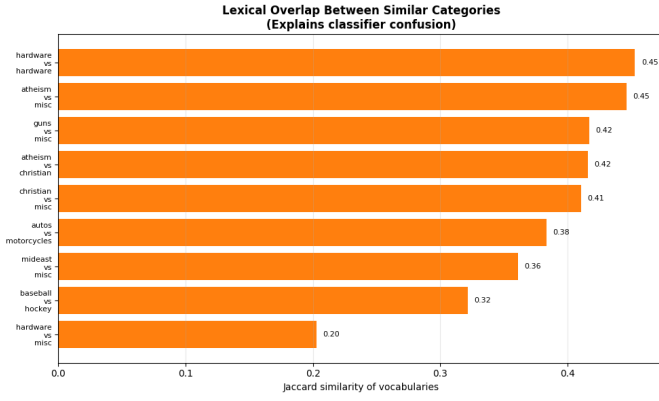


Fig. 5. Jaccard overlap between the vocabularies of similar class pairs. High overlap predicts which classes the models will confuse.

matically related class pairs using the Jaccard coefficient over their word sets. The largest overlaps fall precisely where confusion would be expected on conceptual grounds: the two hardware classes overlap at 0.45, alt.atheism and talk.religion.misc at 0.45, talk.politics.guns and talk.politics.misc at 0.42, alt.atheism and soc.religion.christian at 0.42, and rec.autos and rec.motorcycles at 0.38. This pattern provides an explicit early indicator of what every confusion matrix reported later in the paper confirms: the religion-related, politics-related, and computing-related classes are intrinsically hard to disentangle because they share large portions of their vocabulary. The principal source of error in the subsequent models is therefore already visible in the data itself, before any training has taken place.

V. TASK 1: CLASSICAL FEATURES AND A SMALL NETWORK

The classifier used in this task is deliberately minimal in capacity. It consists of two linear layers separated by a ReLU activation, with a softmax over the twenty target classes at the output. The hidden dimension is set to 256, dropout is applied for regularisation, the learning rate is 10^{-3} , and training proceeds for 30 epochs. Because the network is held fixed across the three sub-tasks, any difference in performance is attributable to the input representation rather than to the classifier, which is the central design choice of the task.

A. Word2Vec embeddings and how they change with training

For part (b) a Word2Vec model is trained on the corpus and each document is represented by the average of its word vectors [13], [17]. The brief poses a useful diagnostic question here: do the learned vectors actually change with additional epochs of training, and if so, with what effect? To address this, the embedding space is examined after 1, 5, and 20 epochs and compared across the three snapshots.

Figure 6 provides the most direct evidence. It presents pairwise cosine-similarity matrices for a fixed set of common words after 1, 5, and 20 epochs. After only one epoch

the space is highly undifferentiated and almost every pair of words exhibits high similarity, visible as a broadly red region. As training progresses to five and then twenty epochs, off-diagonal similarities drop substantially while semantically related words remain close. The computing terms (computer, software, linux, hardware) retain high mutual similarity, the religion terms (god, church, religion, bible, christian) form a coherent block, and the space-related terms (space, nasa, orbit, rocket, launch) cluster together. Training therefore does not merely shift the vectors numerically; it gradually transforms a poorly structured space into one with clearly identifiable topical neighbourhoods.

Figure 7 corroborates this view from a different angle, using a two-dimensional projection of the vectors. The configuration of the same set of common words differs visibly at 1, 5, and 20 epochs, indicating that the points continue to move as training proceeds. To quantify the magnitude of this movement, Figure 8 reports the average cosine distance of the vectors from their one-epoch positions. The mean drift grows from about 0.27 at five epochs to about 0.31 at twenty epochs, indicating that the vectors continue to evolve beyond five epochs but at a diminishing rate, which is consistent with the qualitative picture provided by the similarity matrices.

Using these embeddings, the network attains a test accuracy of 67.2% and a macro F1 of 0.659. The training curves in Figure 10 are well behaved: training and validation accuracy rise together and plateau near 0.71, with only a small final gap. This indicates that the averaged Word2Vec features are informative but limited in capacity, which is consistent with the observation that averaging word vectors over a document tends to dilute the strong individual keywords that are often decisive for topic classification.

B. TF-IDF features and a comparison

For part (c) the embeddings are replaced by TF-IDF vectors while the network is held fixed. Accuracy increases to 68.9% with a macro F1 of 0.678, exceeding the averaged Word2Vec representation. The explanation is closely related to the limitation noted above: newsgroup topics are frequently signalled by a small number of highly discriminative keywords, such as “nasa” or “encryption”, and TF-IDF preserves these terms with their full weight rather than averaging them into a single document vector.

The TF-IDF training curves in Figure 11, however, reveal a clear pathology. Training accuracy rises to approximately 0.97 while validation accuracy peaks around epoch five and slowly declines thereafter, which is a textbook signature of overfitting. The TF-IDF input space contains several thousand sparse features, and the network has sufficient capacity to memorise the training distribution. The representation is therefore strong, but generalisation could likely be improved through early stopping or stronger regularisation. This observation motivates the next sub-task directly.

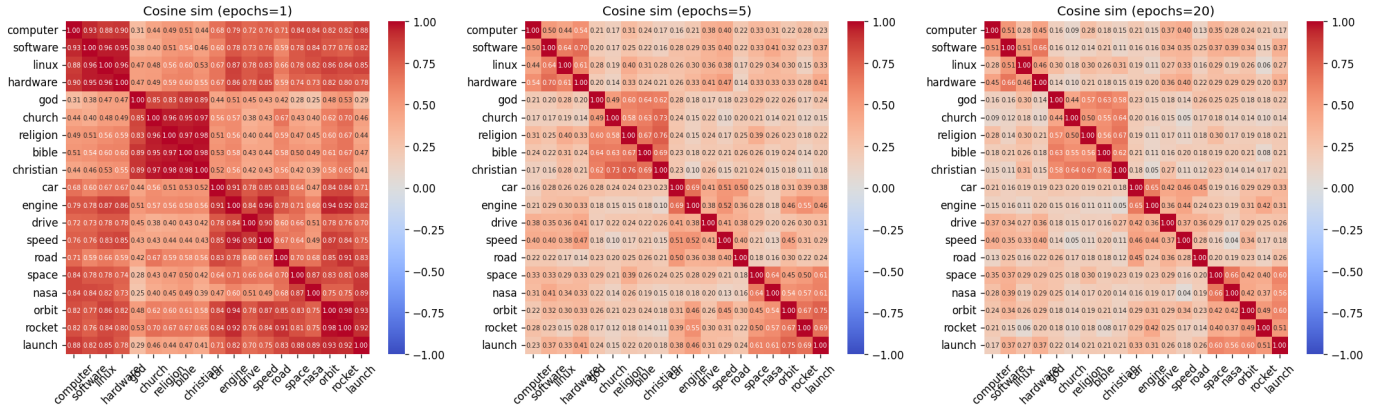


Fig. 6. Cosine similarity between selected words after 1, 5, and 20 training epochs of Word2Vec. At epoch 1 almost everything is similar to everything (wide red region). With more training the space becomes sharper, and only words that share a topic keep a high similarity.

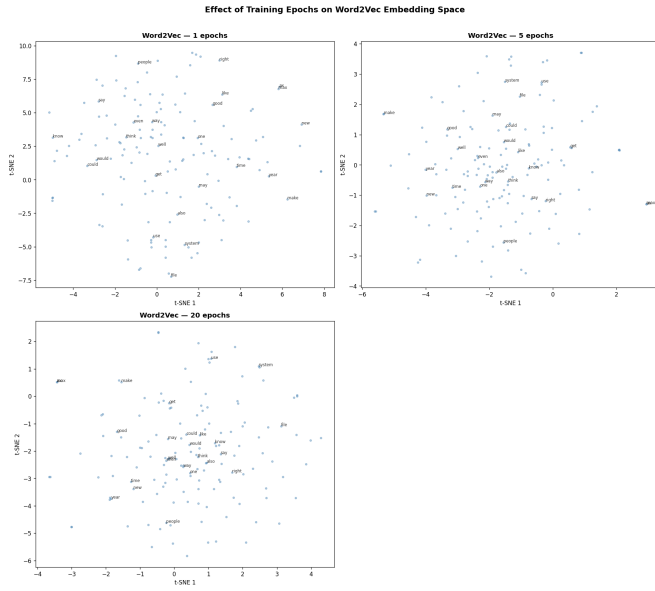


Fig. 7. 2-D projection of the Word2Vec space after 1, 5, and 20 epochs. The arrangement of the same common words changes with training, which shows that the embedding space is reorganising itself.

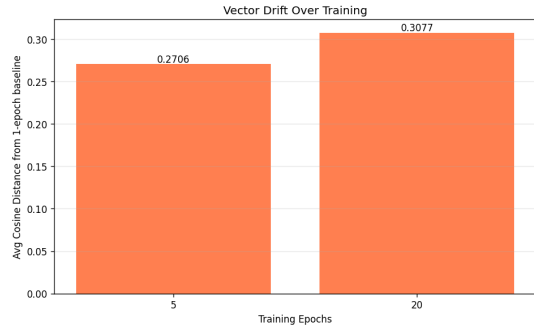


Fig. 8. Average cosine distance of the word vectors from their one-epoch positions. The space keeps changing as I train longer, but the rate of change slows down.

C. An experiment to improve the result

For part (d) a further experiment is required that improves performance while leaving the classifier unchanged. I therefore combine the two representations by concatenating the dense Word2Vec vector and the sparse TF-IDF vector into a single input. The motivation is that the two signals are complementary: TF-IDF captures the strongest discriminative keywords, while averaged Word2Vec contributes a notion of distributional similarity for posts that use different but related vocabulary.

The combined model attains 70.2% accuracy with a macro F1 of 0.687, the strongest result among the three classical configurations. The confusion matrix in Figure 9 indicates where this accuracy is concentrated and where it is not. Topically distinctive classes such as hockey, baseball, and for-sale posts are handled reliably. The principal error region consists of the overlapping groups: `talk.politics.misc` is frequently confused with the other politics classes, and `talk.religion.misc` between atheism and Christianity. These confusions correspond closely to the lexical overlaps identified earlier and recur in every later model in the study.

VI. TASK 2: FINE-TUNING BERT

Task 2 moves from hand-crafted features to a pre-trained contextual encoder. I fine-tune `bert-base-uncased` on the training set, which updates all parameters of the model for the target classification task [15]. The configuration uses a maximum sequence length of 256 tokens, a batch size selected for the available GPU, four epochs of training, a learning rate of 2×10^{-5} , and the AdamW optimiser. The validation set is used to monitor training.

BERT attains 71.1% test accuracy with a macro F1 of 0.698, which is the strongest result obtained up to this point and exceeds every classical baseline. The improvement is attributable to BERT’s ability to model words in bidirectional context, so that the same surface token can carry different meanings in different posts, an effect that neither a bag-of-words representation nor an averaged embedding can capture.

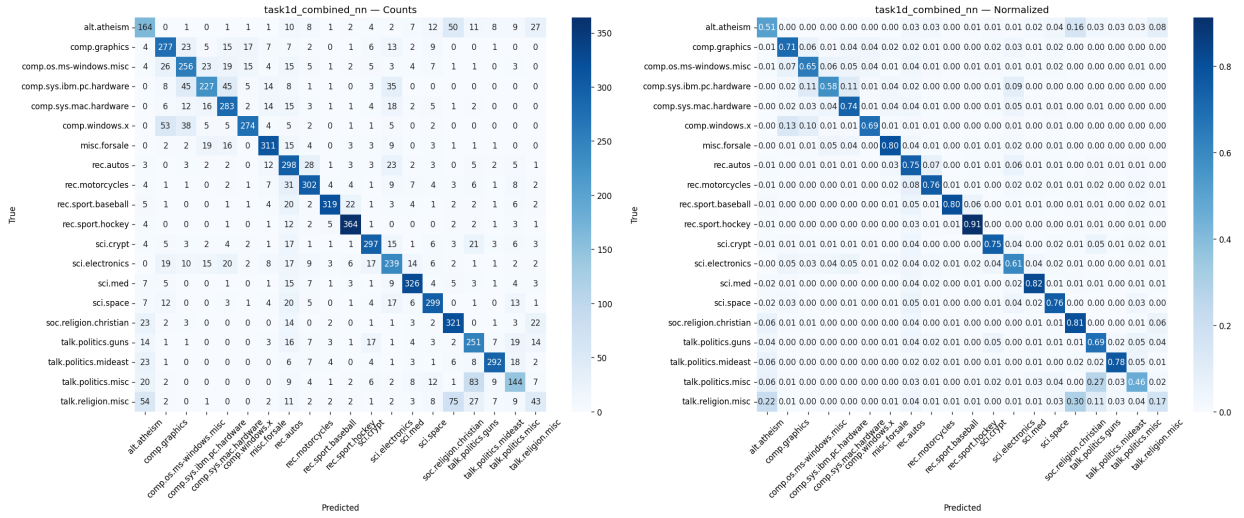


Fig. 9. Confusion matrix for the combined Word2Vec and TF-IDF model (counts on the left, row-normalised on the right). The diagonal is strong for clear topics, but the religion and politics groups bleed into one another.

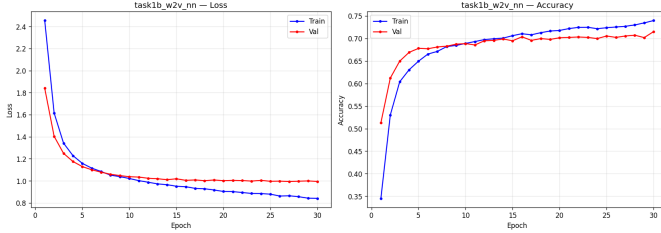


Fig. 10. Training and validation curves for the Word2Vec model. The two lines stay close, which means little overfitting, and accuracy settles near 0.71.

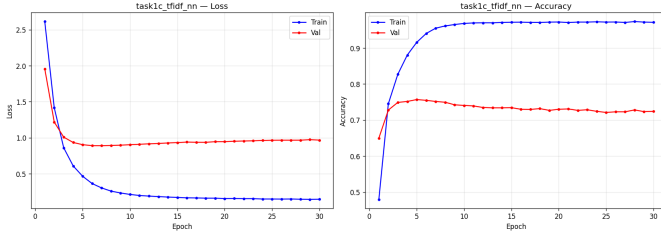


Fig. 11. Training and validation curves for the TF-IDF model. Training accuracy keeps rising while validation peaks early and then falls, which is a clear sign of overfitting.

The confusion matrix in Figure 12 exhibits a strong diagonal, with the residual errors concentrated in the religion and politics block.

A secondary observation concerns compute. An earlier configuration of the same model, trained on a laptop CPU with a short sequence length and only one epoch, reached only 64.2%. The same model trained on a GPU with a longer sequence length and additional epochs reached 71.1%, an improvement of approximately seven percentage points. This gain is not the result of a stronger architecture but of providing the same architecture with sufficient sequence length and training budget. It is a useful reminder that compute and basic

hyper-parameter choices can be as consequential as the choice of model itself.

VII. TASK 3: DOMAIN ADAPTATION WITH MASKED LANGUAGE MODELING

Task 3 asks whether classification performance can be improved by first exposing BERT to in-domain text. The procedure consists of two stages. In the first stage, BERT is further pre-trained on the unlabelled newsgroup corpus with masked language modelling, in which 15% of the tokens are randomly masked and predicted from context. This stage uses only the text and no labels. In the second stage, the adapted model is fine-tuned on the labelled classification task in the same manner as in Task 2.

The adaptation stage is effective. Masked-language-modelling perplexity decreases from approximately 15.9 to approximately 11.2, indicating that the model becomes substantially better calibrated to newsgroup vocabulary and style [16]. After fine-tuning, the adapted model reaches 71.5% accuracy and a macro F1 of 0.694, marginally exceeding plain BERT and making it the strongest supervised model in this study. The magnitude of the gain is modest, which is expected at this dataset scale, but it is consistent and in agreement with prior work on domain-adaptive pre-training [4]. The confusion matrix in Figure 13 resembles that of BERT, with slightly cleaner behaviour on a small number of science and sports classes.

VIII. TASK 4: LLAMA-3 ZERO-SHOT AND FEW-SHOT

Task 4 adopts a fundamentally different approach. Rather than training a model, I query a large instruction-tuned language model, Llama-3 with eight billion parameters, directly on the classification task. The model is loaded in 4-bit form so that it fits within a single GPU’s memory. A system message lists the twenty target categories and instructs the model to respond with exactly one of them. In the zero-shot setting the

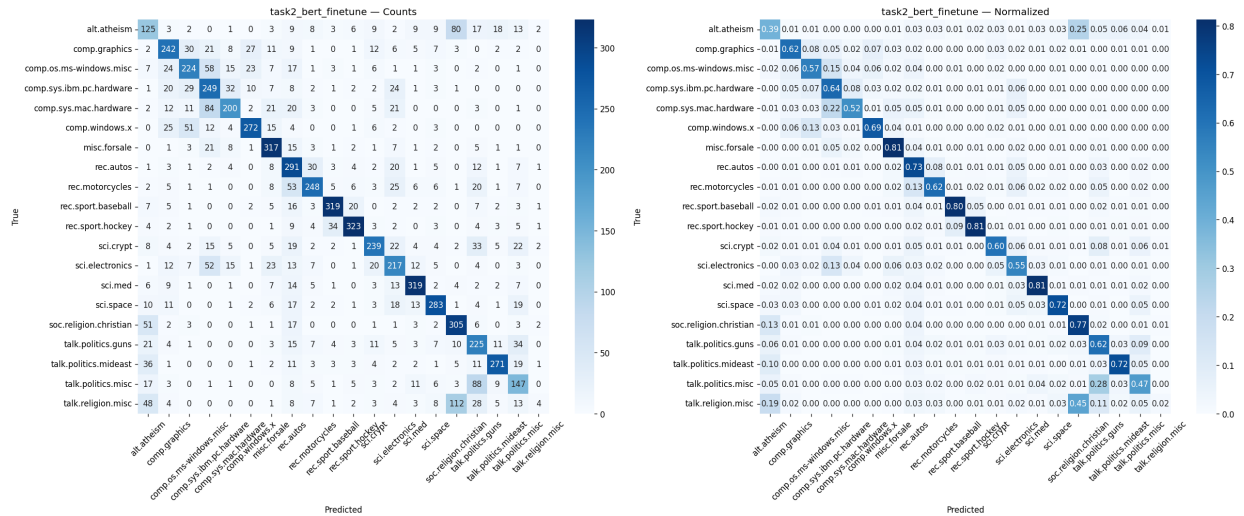


Fig. 12. Confusion matrix for fine-tuned BERT. The model is accurate on most topics. As before, the religion and politics classes are the main source of errors.

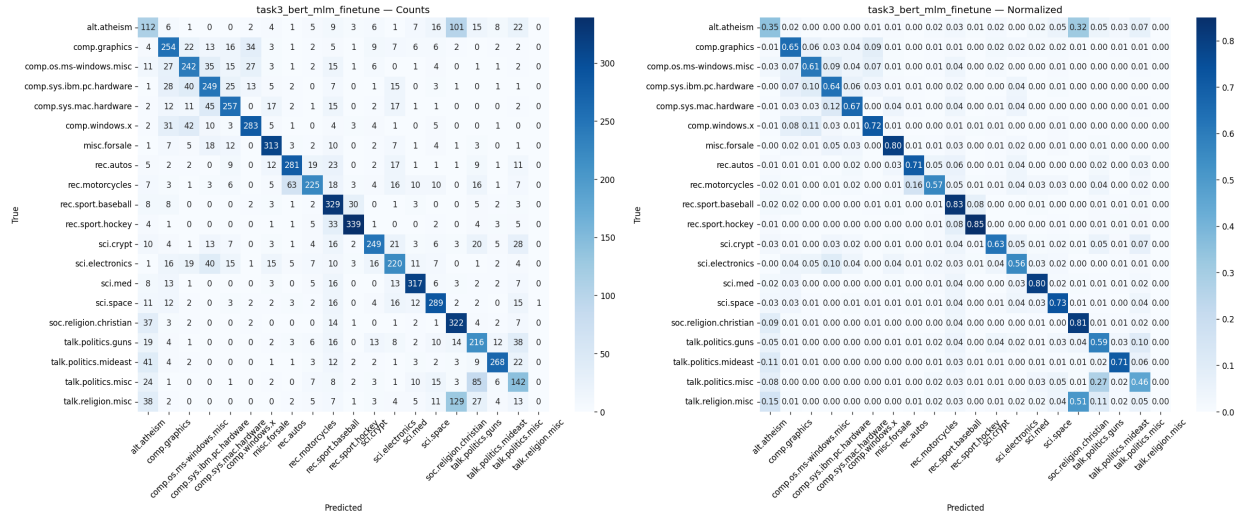


Fig. 13. Confusion matrix for BERT after masked language model domain adaptation and then fine-tuning. The pattern is close to plain BERT, with a small improvement overall.

system message is followed by the test post alone; in the few-shot setting three labelled example posts are inserted before it. To keep inference tractable, all Llama-3 methods are evaluated on a fixed sample of 200 test posts, ensuring that every variant is judged on identical inputs.

Zero-shot prompting attains 51.5% accuracy and a macro F1 of 0.475. This figure is well below BERT, yet it remains notable, since the model has not been trained on this task and nevertheless exceeds the random baseline by an order of magnitude across twenty classes. The confusion matrix in Figure 14 indicates that the model is comparatively reliable on topically distinct categories but tends to default to a small number of broad categories when uncertain.

A more surprising finding is that few-shot prompting performs worse than zero-shot, reaching 47.5% accuracy and a

macro F1 of 0.420. Two factors account for this. The first, and more impactful, is the prompt format. An initial version of the prompt used a plain-text template, which produced approximately 22.5% unparseable outputs because the model either rambled or echoed parts of the prompt. Switching to Llama-3’s native chat template, which wraps system and user messages in the special format used during the model’s instruction tuning, reduced the rate of unparseable outputs to approximately one percent and accounted for the largest improvement in Task 4. The second factor is that three randomly chosen examples cannot adequately represent twenty classes; they bias the model toward the categories shown and add noise. Given that the instruction was already explicit, the examples were on balance harmful. Figure 15 reports the corresponding confusion matrix.

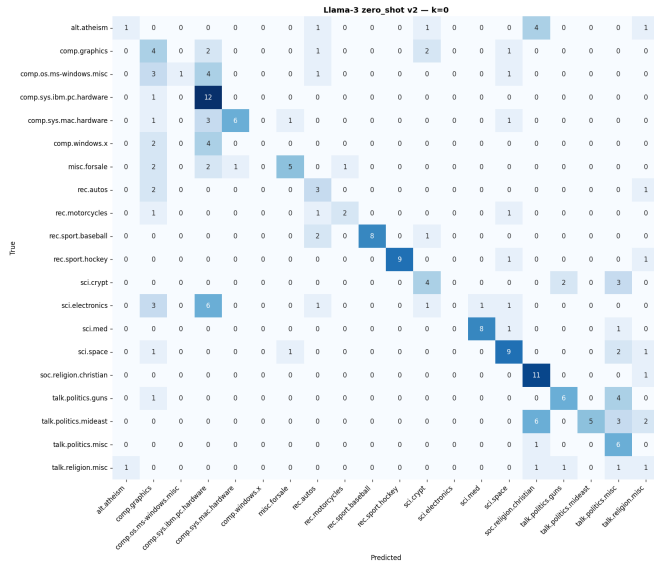


Fig. 14. Confusion matrix for Llama-3 zero-shot, after I fixed the prompt format. The model handles clear topics well and struggles with the overlapping ones.

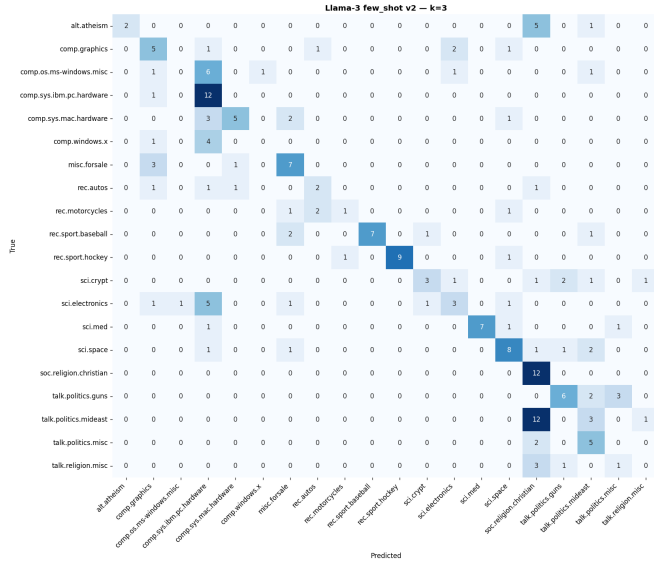


Fig. 15. Confusion matrix for Llama-3 few-shot with three examples. Adding examples did not help, and the model leans toward the few categories it was shown.

IX. BONUS: FINE-TUNING LLAMA-3 WITH QLoRA

The prompting experiments demonstrate that Llama-3 contains useful knowledge but does not exploit it effectively without adaptation. The optional bonus task therefore investigates a simple question: how much can be gained from adapting the model in a computationally inexpensive way? I apply QLoRA, in which the base model is kept frozen in 4-bit form and small low-rank adapter matrices, with rank 16 and scaling factor 32, are inserted into the attention and feed-forward layers. Only approximately 0.2% of the parameters are trained. The adapter is fine-tuned on a class-balanced subset of 3,000 documents,

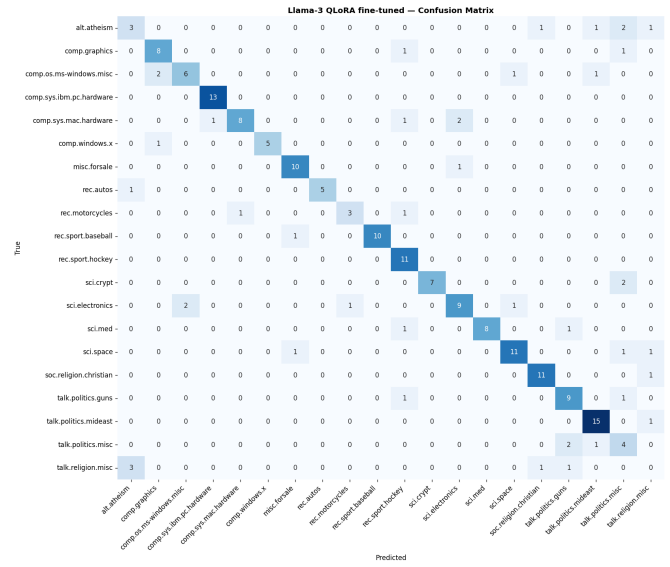


Fig. 16. Confusion matrix for Llama-3 fine-tuned with QLoRA on the 200-post test sample. The diagonal is much stronger than in the prompting results, and most topics are now handled well.

150 per class, drawn from the training partition. The task is reformulated as an instruction in which the model is required to emit a category name, and the loss is computed exclusively on the answer tokens so that the model learns to produce the label rather than to reproduce the prompt. Training uses a learning rate of 2×10^{-4} and the paged 8-bit AdamW optimiser.

The result is the highest measured in the entire study. QLoRA reaches approximately 78.0% accuracy on the 200-post evaluation sample, exceeding both prompting baselines and the BERT-based models on that sample. The confusion matrix in Figure 16 exhibits a clean and pronounced diagonal. Categories on which prompting consistently failed, such as hardware, the Middle East politics class, and Christianity, are now predominantly correct. Residual errors are concentrated in the two least specific classes, talk.politics.misc and talk.religion.misc, which is unsurprising since these are intrinsically ambiguous even for human readers. Adapting only a small fraction of the parameters of a large model is therefore sufficient to convert a weakly performing prompted system into the highest-scoring model in this comparison.

A caveat must be acknowledged regarding the evaluation protocol. The Llama-3 results, including QLoRA, are measured on the 200-post sample, whereas the classical and BERT models are measured on the full 7,532-document test set. Consequently, the 78% and the 71.5% figures are not strictly comparable on identical inputs. The QLoRA result is nonetheless a strong signal, and a fully matched comparison on the complete test set is identified as future work.

X. OVERALL COMPARISON AND DISCUSSION

Table I consolidates the results of every method. Reading the table from top to bottom traces the trajectory of the project

from classical features to parameter-efficient large-model fine-tuning.

Several conclusions follow from the comparison. The first concerns feature combination: the dense Word2Vec and sparse TF-IDF representations are individually incomplete, and their concatenation produces a stronger classifier than either representation alone. The second concerns context: BERT, which models words in bidirectional context, outperforms every classical configuration despite using the same labelled data. The third concerns domain adaptation: continued masked-language-modelling pre-training on in-domain text yields a small but consistent improvement over plain BERT and a measurable reduction in perplexity, indicating that the adaptation is real rather than illusory.

The fourth, and most interesting, conclusion concerns the large language model. Used by prompting alone, Llama-3 performs poorly relative to a fine-tuned BERT and is in fact harmed by the addition of random in-context examples, because the format of the prompt has a far stronger effect on accuracy than the number of examples does. Switching to the model’s native chat template was the single most impactful change throughout Task 4. Once the model is adapted, however, even by training only a tiny fraction of its parameters through QLoRA, it becomes the highest-scoring method in the comparison. The headline of the project is therefore not that the largest model always wins, but that a large model is competitive only in conjunction with an appropriate adaptation strategy: under prompting alone it trails a small network, while under fine-tuning it leads the field.

TABLE I
TEST RESULTS FOR EVERY METHOD. THE LLAMA-3 METHODS WERE MEASURED ON A FIXED 200-POST SAMPLE; THE OTHERS ON THE FULL TEST SET.

Method	Accuracy	Macro-F1	Train data
Word2Vec + NN	67.2%	0.659	9,051
TF-IDF + NN	68.9%	0.678	9,051
Combined + NN	70.2%	0.687	9,051
BERT fine-tune	71.1%	0.698	9,051
BERT + MLM	71.5%	0.694	9,051 + 17,886
Llama-3 zero-shot	51.5%	0.475	0
Llama-3 few-shot ($k=3$)	47.5%	0.420	3 examples
Llama-3 QLoRA	78.0%	0.746	3,000

A final observation cuts across every model in the study. The religion-related classes and the politics-related classes overlap heavily in vocabulary and consequently overlap in predictions. This pattern was already visible in the lexical analysis of the corpus, prior to any training, and persists in every confusion matrix reported in this paper. It is therefore a property of the dataset rather than a deficiency of any individual model.

XI. LIMITATIONS AND FUTURE WORK

Several limitations of this study point naturally to further work. The most important is the evaluation gap noted above: the Llama-3 methods are evaluated on a 200-post sample,

whereas the other methods use the full test set. A like-for-like comparison would require evaluating QLoRA on the entire test set, which is the most direct next step. A further consequence of the small evaluation sample is that the confidence intervals on the Llama-3 accuracy figures are correspondingly wider, so the apparent margin between QLoRA and the BERT models should be interpreted with this caveat in mind.

The TF-IDF and combined models exhibit clear overfitting, so stabilising the baseline with explicit regularisation—a higher dropout rate, L_2 weight decay on the network parameters, and early stopping on the validation loss—would likely yield further gains in validation accuracy. The few-shot examples are sampled at random; replacing this with a similarity-based selector—for example, a k -nearest-neighbour search over the Word2Vec or BERT embeddings to retrieve the most relevant examples for each query post—would likely make few-shot helpful rather than harmful. For the BERT pipeline, stronger encoders such as RoBERTa or longer in-domain pre-training could plausibly raise the supervised ceiling further. More broadly, since the religion- and politics-related classes constitute the dominant source of error across every model, a targeted approach that distinguishes specifically among these closely related categories would benefit every model in the comparison simultaneously. Finally, scaling the QLoRA experiment to more training data and more epochs is a natural further direction.

XII. CONCLUSION

The objective of this project was to solve a single text-classification task with four families of methods and to interpret the resulting hierarchy rather than merely to report it. Accuracy increased progressively with the addition of task-specific training, from approximately 67% for classical features to approximately 71% for fine-tuned and domain-adapted BERT. An untrained large language model used solely through prompting reached only about 51%, whereas the same model fine-tuned at low cost via QLoRA reached approximately 78% on the matched 200-post evaluation sample. This last figure is suggestive rather than conclusive, since it is not measured on the full test set, but it indicates that parameter-efficient fine-tuning of the LLM is at least competitive with full BERT fine-tuning at this dataset scale. The overarching conclusion is that method selection must be matched to data scale and available compute: a small network with carefully engineered features is a sound baseline, fine-tuned BERT is a robust default, and a large language model is only as effective as the adaptation procedure applied to it. The study further confirms that domain adaptation yields a small but reliable gain and that, for a strongly instruction-tuned model, the structure of the prompt can outweigh the number of in-context examples.

XIII. CODE AND DATA AVAILABILITY

All of the code, the documentation, and this report are available in a public GitHub repository, which has been shared with the lecturer for evaluation:

<https://github.com/ProNilabh/nalapro-project>

The final report is located in the repository at `docs/report.pdf`. The dataset is not included in the repository, as required; it is downloaded directly through scikit-learn’s `fetch_20newsgroups` loader. Every experiment, with its settings, training curves, and final metrics, was logged to Weights and Biases and can be inspected at <https://wandb.ai/nilabhpandey7-hslu/nalapro-project>.

XIV. GENERATIVE AI DECLARATION

In line with the course policy, I declare that generative AI tools were used during this project. They served as an assistant for writing and debugging code, for clarifying unfamiliar aspects of libraries such as Hugging Face Transformers and PEFT, and for improving the clarity and wording of this report. I did not use AI to generate my results. The design of the experiments, the choice of models and hyper-parameters, the training runs, and the analysis of the outputs are my own work. Every result reported here was produced by my own code and logged to Weights and Biases. I have reviewed and understood all of the code and the content of this report, and I am able to explain and defend each part of it.

REFERENCES

- [1] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [2] T. Mikolov, K. Chen, G. Corrado, and J. Dean, “Efficient estimation of word representations in vector space,” in *Proc. ICLR Workshop*, 2013.
- [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [4] S. Gururangan et al., “Don’t stop pretraining: Adapt language models to domains and tasks,” in *Proc. ACL*, 2020, pp. 8342–8360.
- [5] T. Brown et al., “Language models are few-shot learners,” in *Proc. NeurIPS*, 2020, pp. 1877–1901.
- [6] A. Grattafiori et al. (Meta AI), “The Llama 3 herd of models,” [arXiv:2407.21783](https://arxiv.org/abs/2407.21783), 2024.
- [7] E. J. Hu et al., “LoRA: Low-rank adaptation of large language models,” in *Proc. ICLR*, 2022.
- [8] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized LLMs,” in *Proc. NeurIPS*, 2023.
- [9] F. Shahbazi, “NLP Pipeline,” Lecture notes for NALAPRO, Dept. of Information Technology, Lucerne Univ. of Applied Sciences and Arts (HSLU), Lucerne, Switzerland.
- [10] F. Shahbazi, “Tokenization,” Lecture notes for NALAPRO, Dept. of Information Technology, HSLU, Lucerne, Switzerland, 2026.
- [11] F. Shahbazi, “Explicit Semantic Analysis,” Lecture notes for NALAPRO, Dept. of Information Technology, HSLU, Lucerne, Switzerland.
- [12] F. Shahbazi, “N-Gram Detection (Frequency-based),” Lecture notes for NALAPRO, Dept. of Information Technology, HSLU, Lucerne, Switzerland.
- [13] F. Shahbazi, “N-Gram Detection (Prediction-based) — Word2Vec,” Lecture notes for NALAPRO, Dept. of Information Technology, HSLU, Lucerne, Switzerland.
- [14] F. Shahbazi, “Transformer, Pretraining, BERT,” Lecture notes for NALAPRO, Dept. of Information Technology, HSLU, Lucerne, Switzerland, 2026.
- [15] F. Shahbazi, “Fine-tuning Models: Classification,” Lecture notes for NALAPRO, Dept. of Information Technology, HSLU, Lucerne, Switzerland, 2026.
- [16] F. Shahbazi, “Text Generation,” Lecture notes for NALAPRO, Dept. of Information Technology, HSLU, Lucerne, Switzerland, 2026.
- [17] F. Shahbazi, “Creating Text Embedding Models,” Lecture notes for NALAPRO, Dept. of Information Technology, HSLU, Lucerne, Switzerland, 2026.